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# Modeling the railway network design problem: A novel approach to considering carbon emissions reduction

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ABSTRACT

In recent years, air pollution has become a severe environmental problem. Many countries are seeking effective methods to reduce carbon emissions. Among the sources of air pollution, the transportation industry is a large one. Thus, developing low carbon emissions transportation is a new trend. Since railway transport possesses the advantages of high volume and low emissions, a new idea for transportation development is to divert truck freights to railway by improving the capacity of the railway. Based on this situation, this paper analyzes the carbon emissions of railway and highway transport and estimates the environmental benefit of building a railway. Next, a bi-level programming model for railway network design is recommended. Different from the classical model, the opportunity cost and carbon emissions of the infeasible flows, which cannot be transported by railway and will be served by highway, are considered in the objective function, enabling the model to balance the investment and carbon emissions. To comply with the practice of railway planning, we introduced the method to deal with passenger flow, empty car flow and relationships among the links in a corridor. In the end, a numerical example is conducted to test the model and compare the final decisions when different weights are set on the carbon emissions.

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## 1. Introduction

As the problem of air pollution is becoming increasingly severe, carbon emissions reduction has attracted much attention from the international community.

Among the many sources of carbon emissions, the transportation industry is a large one. Thus, promoting the development of low-carbon transport will greatly benefit the world's carbon emissions reduction. Compared to highway transport, railway and water transport are better choices for developing green transportation because of their low energy consumption and carbon emissions.

Therefore, in 2011, the European Union issued a white paper, titled "Roadmap to a Single European Transport Area – Towards a competitive and resource efficient transport system," stating that 30% of road freight over 300 km should shift to other modes such as rail or waterborne transport by 2030, and more than 50% by 2050 (European Commission, 2011).

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At the end of 2015, the Ministry of Transport of China also recommended that railways and waterways should undertake more freight transportation.

Since the capacity of inland waterways is usually limited due to flood or drought, diverting some truck freight to the railway is a wise choice.

Take China as example, in 2015, the total freight transport completed by the 102 thousand kilometer railway (not including the high-speed railway) was 2375.43 billion ton-kilometers. The total freight transport completed by the 4577 thousand kilometer highway reached 5795.57 billion ton-kilometers, which was 1.44 times greater than railway transport.[1] Based on data from the Ministry of Transport of China, Fig. 1 provides the last five years of $CO_2$ emissions of railway transport and highway transport.

An analysis of the data in Fig. 1 reveals that the average $CO_2$ emissions per workload of highway transport is 3.36–4.64 times more than that of railway transport. Obviously, railway transport is more environmentally friendly. However, due to the capacity shortage of some railway corridors, hundreds of millions of tons of bulk freight (such as coal) is transported on highway for long distance, leading to significant consumption of high-level energy sources (such as diesel) and carbon emissions.

Fig. 1 shows that the total $CO_2$ emissions of highway freight transport is 7.89–10.45 times more than that of railway freight transport. Such a situation of these two modes is irrational because some freight that should have been transported by train is transported by truck. Therefore, adjusting the share ratio of the two modes and diverting a portion of freight flows from highway to railway will reduce the carbon emissions of the transportation system efficiently. Based on the $CO_2$ emissions per workload in 2015, if 1 ton-kilometer workload can be diverted from highway to railway, the $CO_2$ emissions will reduce 38.87 g.

Thus, if 10% of the highway transport workload (approximately 579.6 billion ton-kilometers) can be shifted to the railway, carbon emissions will reduce 22.53 million tons per year. And if 30% (about 1738.8 billion ton-kilometers) is shifted, 67.58 million tons carbon emissions will be reduced. This will make a great contribution to the world's carbon emissions reduction.

The issue is whether the railway can manage the highway's freight.

In previous decades, the capacity of China's railway has always been insufficient. The capacity shortage was not mitigated until a lot of high-speed railway was built, and some passengers were attracted to high-speed railways. However, there is little capacity surplus, especially the corridors for coal transport. Currently, the average load of the railway is still as much as 23 million tons per kilometer per year, which is higher than the load of most other countries.[2] Some railway corridors are even burdened much more than the average load. For example, the Da-Qin railway in northern China, which is known as the most famous coal-transport corridor, reached approximately 400 million tons in one year.[3]

Therefore, in order to reduce carbon emissions by diverting freight flow from highway to railway, optimizing the structure of railway network is a scientific problem that needs to be addressed urgently both in theory and practice. This is what we focus on in our paper.

## 2. Literature review

Railway network planning can be abstracted as network design problem (NDP), which is a classical problem in the field of traffic and transportation that many scholars have researched. Among the branches, the research for urban traffic network started relatively early. Leblanc (1975) addressed the problem of determining which links should be improved in an urban road network so that total congestion in the city is minimized. Later, many researchers were devoted to the discrete and continuous NDP for road transportation. Some of them focused on the problem of whether a link should be constructed, and the number of lanes that the constructed links should have (Drezner and Wesolowsky, 2003; Wang et al., 2013). Others aimed at determining the optimal capacity expansions of the links by using continuous decision variables (Liu and Wang, 2015; Wang et al., 2015). The research on railway network design problem (RNDP) started relatively late. Kuby et al. (2001) presented that railway projects can be built all at once or in stages and developed a heuristic time sequence procedure with a cost adjustment factor to solve the "project staging" problem. By removing all the small traffic demands below a certain cutoff level from the OD matrix and preloading them onto their shortest paths, the model size was reduced significantly while very little was lost in the way of realism and optimality. Lin et al. (2002) addressed the optimization decision for the RNDP to determine where a new railway line should be built, or where an existing railway line should be strengthened based on the project investment and cost of routing cars. Yan et al. (2007) discussed the impact of limited investment on the railway project's choice and developed a model of RNDP in several plan stages.

In terms of the research on multimodal network, Arnold et al. (2004) dealt with the problem of optimally locating rail/road terminals for freight transport. A linear 0-1 program was formulated and solved by a heuristic approach. Janic (2007) developed a model for calculating comparable combined internal and external costs of intermodal and road freight transport networks. This would be very helpful for transportation network design. Pazour et al. (2010) put forward that highway congestion could be reduced by diverting freight traffic to the high-speed railway. Thus an un-capacitated network

[1] The data are cited from National Bureau of China (http://data.stats.gov.cn/easyquery.htm?cn=C01).

[2] According to the data from the National Bureau of Statistics of China, the total freight transport completed by the 102 thousand kilometer railway (not including the high-speed railway) was 2375.43 billion ton-kilometers. Thus, in 2015, the average load of the railway line in one year is approximately 23 million tons per kilometer.

[3] The data is cited from the Compilation of National Railway Statistics of China.(2015).

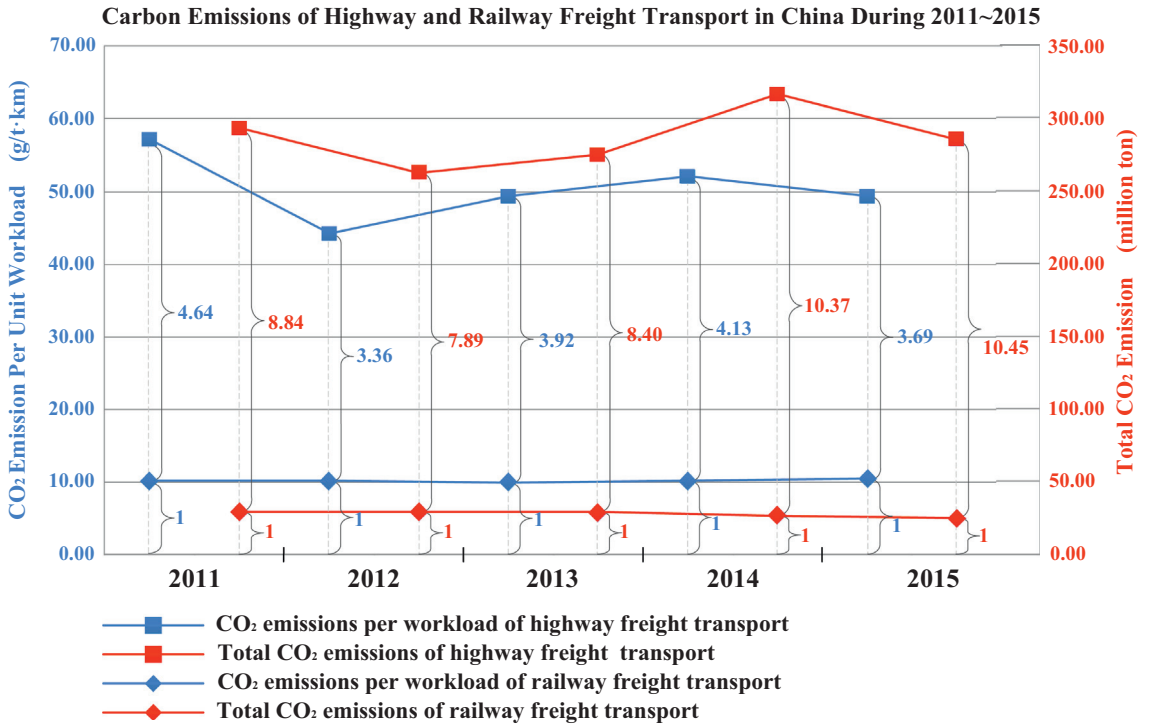



**Fig. 1.** Carbon emissions of highway and railway freight transport in China during 2011–2015 (The $CO_2$ emissions are calculated according to the energy consumption data of road and railway from Ministry of Transport of China (http://zizhan.mot.gov.cn/zfxxgk/bnssj/zhghs/201605/t20160506_2024006.html)).

design model considering highway and high-speed railway was developed. The model aimed to minimize the total travel time, and for the capacity constraint, a post-processing step was designed. It is worthy of pointing out that the idea presented in this paper is also helpful in the development of low-carbon transportation. By exploring the multi-period NDP of comprehensive transport, Sun et al. (2011) proposed a bi-level programming model with uncertain demand. The model can simultaneously optimize the final form of the comprehensive transport network and the timing of construction.

Besides, many scholars made in-depth study on methodologies of NDP. Magnanti and Wong (1984) provided a unifying view for synthesizing many network design models and showed that many of the most celebrated combinatorial problems that arose in transportation planning are specializations and variations of a generic design model. Several researchers formulated NDP as bi-level programming model. Gao et al. (2005) proposed a new solution algorithm by using the support function concept to solve bi-level programming model for the discrete network design problem. However, some scholars reformulated the model as single-level programming model before designing the corresponding algorithm (Frangioni and Gendron, 2009; Wang and Lo, 2010; Farvaresh and Sepehri, 2011; Fontaine and Minner, 2016). In recent years, many studies focused on the innovation of algorithm for NDP. Katayama et al. (2009) proposed a capacity scaling heuristic using a column generation and row generation technique to address the multi-commodity capacitated network design problem. Then Benders decomposition algorithm was adopted to solving NDP (Gendron, 2011; Fakhri and Ghatee, 2016).

According to the above analysis, the previous studies mainly focused on reducing travel cost of network users, simplifying the models and designing algorithms for solving the problem. These researchers made significant contributions, both in theory and practice, which are useful in subsequent research. However, the objectives that the publics and governments are pursuing have changed. Since green transport is becoming a dominant goal throughout the world, improving railway network have already become an important measure to adjust the ratio of railway and highway capacity and to reduce carbon emissions. Therefore, carbon emissions should be considered when planning the railway network. Nevertheless, as far as we know, few of the existing literatures which focus on railway network planning take carbon emissions into consideration.

Thus, based on prior research, this paper makes the following improvements by considering the current background and the feature of railway corridor. Firstly, we assume that the freight flows that cannot be served by railway will all be assigned to the highway. The carbon emissions of these flows are considered in the objective function of the model. In this way, we can balance the investment and carbon emissions. Secondly, since the distribution of empty cars also occupies considerable railway capacity, we modified the O-D matrix by adding virtual freight flows that were equivalent to the distribution of empty cars, making the matrix reflect the actual traffic volume. Thirdly, since the links in a certain railway corridor are usually at the same level, the same project should be implemented on each of them. For example, all the links should be double-track, or all should be electrified. We formulated this factor and the model size was reduced.

The remainder of this paper is organized as follows. Section 3 develops the bi-level programming model of RNDP considering reducing carbon emissions. Section 4 introduces the railway freight flow assignment models for lower-level. Section 5 recommends three practical factors that need to be taken into consideration when designing a railway network. A numerical example is conducted in Section 6. Finally, the study's conclusions are discussed in Section 7.

## 3. Mathematical formulation for the RNDP

### 3.1. Problem description

The process of railway network planning can be roughly divided into the following three steps in China.

(1) Collecting the candidate projects. There are mainly two sources of candidate projects. One is from local governments and the other is from related research institutes. The local governments propose their advice of building railway based on the demand of the local development. The institutes propose the candidate projects usually aiming at improving the structure of the railway network.
(2) Selecting appropriate projects. Since the total budget is limited, not all of the candidate projects can be implemented. Therefore, some methods should be adopted to select appropriate projects and form a scheme.
(3) Making decision. The related administration will review the scheme and make the final decision.

Our research focuses on the second step of the process, namely, selecting appropriate projects from given candidate projects.

### 3.2. Notations

When studying the railway network design problem from the perspective of carbon emissions, not only the investment and transportation costs but also the carbon emissions costs caused by highway and railway should be taken into consideration. Therefore, the model includes many parameters and variables. The common used ones are listed as follows (Some parameters or variables are of different forms in different lower-level models. Thus their meanings are explained when they appear.):

| | |
|---|---|
| **Sets:** | |
| $V$ | Set of nodes (stations) in the whole network |
| $V(i)$ | Set of nodes (stations) adjacent to node $i$ |
| $E$ | Set of links indexed by $m$. In practice, trains can run on a railway in either direction, thus link $m$ can be regarded as two arcs with opposite directions, which are denoted by $m'$ and $m''$, respectively. The set of arcs is denoted by $G$. |
| $Q_m$ | Set of candidate projects for link $m$ indexed by $q$ |
| **Parameters:** | |
| $F_{ij}$ | Freight transportation demand from $i$ to $j$ |
| $\beta_{ij}$ | Charge for transporting per unit freight from $i$ to $j$ on railway |
| $c_m^q$ | Unit transportation cost on railway if project $q$ is implemented on link $m$ |
| $\rho$ | Opportunity cost, meaning the potential loss when a railway company loses one ton-km of transportation demand |
| $l_{ij}^{\text{Rail}}$ | Distance of railway chargeable path between $i$ and $j$, chargeable path is a given path for calculating the charge for freight transportation by railway company. We assume that the chargeable path is the shortest path between $i$ and $j$ |
| $l_{ij}^{\text{Road}}$ | Average distance of highway between $i$ and $j$ |
| $l_m$ | Length of link $m$ |
| $\theta^{\text{Rail}}$ | Carbon emissions per ton-km on railway |
| $\theta^{\text{Road}}$ | Carbon emissions per ton-km on highway |
| $\lambda$ | Weight coefficient of carbon emissions |
| $\omega_m$ | Volume of freight flow on link $m$, $\omega_m = \omega_{m'} + \omega_{m''}$ |
| $A_m^q$ | Capacity of link $m$ after implementing project $q$. The capacity here contains both directions |
| $T_m^q$ | Depreciation period of project $q$ on link $m$ |
| $I_m^q$ | Investment of project $q$ on link $m$ |
| $B$ | Budget |
| **Decision variables:** | |
| $y_m^q$ | Binary decision variable which equals 1, if link $m$ chooses project $q$ and equals 0 otherwise |
| $f_{ij}$ | Volume of feasible flow of $F_{ij}$, meaning the freight flow that can be served by railway |
| $u_{ij}$ | Infeasible flow form $i$ to $j$, meaning the freight flow that cannot be served by railway, and all will go on highway |

### 3.3. Uniqueness of choosing project

When designing a railway network, we make decisions about the following three problems under a certain annual budget.[4]

(1) Whether a railway should be built between two given nodes
(2) What type of railway should be built
(3) Which existing railway should be improved

If no railway exists between the two given nodes, we regard link $m$ as a virtual link, and its candidate projects may be a single-track non-electrified railway, a single-track electrified railway, a double-track non-electrified railway, a double-track electrified railway, and so on (In China, the railways are classified into four levels. However, the method for differentiating the levels is very complex. In order to describe the problem intuitively, we differentiate the railways based on their physical configurations). The decision variables for link $m$ can be denoted using the following vector:

$$\boldsymbol{y}_m = (y_m^1, y_m^2, \ldots, y_m^{|Q_m|}) \tag{1}$$

Of course, it is possible for none of the candidate projects to be chosen. This means that the link keep its situation unchanged. For example, a link with a single-track railway will remain a single-track one. If the link remains unchanged, a virtual project which is denoted by $y_m^0$ can be defined. Thus formula (1) can be modified as follows:

$$\boldsymbol{y}_m = (y_m^0, y_m^1, y_m^2, \ldots, y_m^{|Q_m|}) \tag{2}$$

For example, if there is a single-track non-electrified railway between the nodes, the candidate projects for link $m$ may be a single-track electrified railway, a double-track non-electrified railway, and a double-track electrified railway or none of the candidate projects to be chosen. We have

$$\boldsymbol{y}_m = (y_m^0, y_m^1, y_m^2, y_m^3) \tag{3}$$

Obviously, only one of the $|Q_m|$ candidate projects for link $m$ can be chose in one stage. Thus we have the following constraint.

$$\sum_{q \in Q_m} y_m^q = 1 \tag{4}$$

### 3.4. Objective function considering carbon emissions reduction

Since the problem involves shifting freight flow from highway to railway, the O-D matrix contains the transportation demand of railway and highway. When the capital investment is limited, not all the candidate projects can be implemented. Therefore, a planned railway network may not meet all the potential transportation demands. This will result in the occurrence of infeasible flows. We assume that the highway network's capacity is sufficient, and freight that cannot be served by the railway will all be delivered by truck. Thus, the infeasible flow will result in two costs: one is the opportunity cost caused by the loss of customers; and the other is the increase of carbon emissions. These costs can be formulated as follows:

$$Z_1(\boldsymbol{u}) = \sum_{i \in V} \sum_{j \in V,\ j \neq i} (\rho l_{ij}^{\text{Rail}} + \lambda\theta^{\text{Road}} l_{ij}^{\text{Road}}) u_{ij} \Rightarrow min \tag{5}$$

where $\boldsymbol{u}$ represents the vector of infeasible flows, which is defined as follows:

$$\boldsymbol{u} = (u_{12}, u_{13}, \ldots u_{ij} \ldots, u_{|V|(|V|-1)}) \tag{6}$$

Another objective of the railway network design is to reduce the transportation cost and carbon emissions cost in the railway system. These costs can be formulated as follows:

$$Z_2(\boldsymbol{\omega}, \boldsymbol{y}) = \sum_{m \in E} \sum_{q \in Q_m} (c_m^q + \lambda\theta^{\text{Rail}}) \omega_m l_m y_m^q \Rightarrow \ \min \tag{7}$$

where $\boldsymbol{\omega}$ represents the vector of load on all the links, which is defined as follows:

$$\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots, \omega_{|E|}) \tag{8}$$

and $\boldsymbol{y}$ represents the vector of all the decision variables which is defined as follows:

[4] In China, the total investment in railway reached 3500 billion CNYs from 2011 to 2015, namely about 700 billion CNY per year in average (http://finance.sina.com.cn/china/gncj/2016-01-04/doc-ifxneept3676961.shtml).

$$\boldsymbol{y} = (\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{|E|}) \tag{9}$$

The last factor in the objective function is the investment. It is certain that we would like to achieve the benefit while investing as few as possible. Thus, we minimize the annual investment in the following formula:

$$Z_3(\boldsymbol{y}) = \sum_{m \in Eq \in Q_m} \sum \frac{I_m^q}{T_m^q} y_m^q \Rightarrow \min \tag{10}$$

In fact, the annual investment should be calculated according to a discounted rate. For simplicity, we use the average of the total cost as an approximate substitute.

In summary, the objective function of the RNDP's upper-level model can be shown as follows:

$$\min Z_U(\boldsymbol{u}, \boldsymbol{\omega}, \boldsymbol{y}) = Z_1(\boldsymbol{u}) + Z_2(\boldsymbol{\omega}, \boldsymbol{y}) + Z_3(\boldsymbol{y}) \tag{11}$$

In order to illustrate how the consideration of carbon emissions influences the railway network design, a simple case is analyzed as follows:

In Fig. 2, two candidate projects are shown with blue dotted lines. Project I is building new railway ④ → ⑧ and project II is building new railway ④ → ⑤ → ⑥. The length of ④ → ⑧ is obviously shorter, and it usually requires lower costs. Thus, if we assess the projects from the perspective of investment only, project I has an apparent advantage. However, ④ → ⑤ → ⑥ is in parallel with a busy highway. When this line is built, part of the freight flow on highway can be diverted to the railway. This will reduce carbon emissions and bring an environmental benefit. From this perspective, project II possesses a greater advantage. In this situation, the final decision may be changed.

### 3.5. Model formulation

According to the notation and analysis, the RNDP can be formulated as a bi-level programming model as follows:

$$(\mathrm{U0}) \min Z_U(\boldsymbol{u}, \boldsymbol{\omega}, \boldsymbol{y}) \tag{12}$$

$$\text{s.t.} \quad \sum_{q \in Q_m} y_m^q = 1 \quad \forall m \in E \tag{13}$$

$$\sum_{m \in Eq \in Q_m} \sum I_m^q y_m^q \leqslant B \tag{14}$$

$$y_m^q \in \{0, 1\} \qquad \forall m \in E, \ \forall q \in Q_m \tag{15}$$

After a solution is generated by the upper-level, namely a solution $\boldsymbol{y} = (y_1^0, y_1^1, \ldots, y_m^q, \ldots, y_{|E|}^{Q_{|E|}}) = (0, 1, \ldots, 1, \ldots, 0)$ is obtained, $\boldsymbol{u}$ and $\boldsymbol{\omega}$ need to be solved by lower-level as follows:

$$(\mathrm{L0}) \max Z_L(\boldsymbol{u}, \boldsymbol{\omega}, \boldsymbol{x}) \tag{16}$$

$$\text{s.t.} \quad h_1(\boldsymbol{u}, \boldsymbol{f}) = \boldsymbol{0} \tag{17}$$

$$h_2(\boldsymbol{f}, \boldsymbol{x}) = \boldsymbol{0} \tag{18}$$

$$g(\boldsymbol{\omega}, \boldsymbol{y}) \leqslant \boldsymbol{0} \tag{19}$$

In L0, $\boldsymbol{f} = (f_{12}, f_{13}, \ldots, f_{ij}, \ldots, f_{|V|(|V|-1)})$. $\boldsymbol{x}$ is the vector of decision variables for railway freight flow assignment, the meaning and form of $\boldsymbol{x}$ are different in different models. This will be described in detail in Section 4.

The model contains upper-level and lower-level. The upper-level is for choosing projects. It contains three constraints: (1) only one project can be chosen for each link; (2) the total investment should be within the budget; and (3) the decision variables can be either 0 or 1.

The lower-level focuses on railway freight flow assignment (RFFA). Its objective is to maximize the railway's profit. In general, all RFFA which consider infeasible flows contain three basic constraints: (1) the flow conservation of feasible flows and infeasible flows; (2) flow conservation on the path or at the node; and (3) capacity limitation. The detailed RFFA will be discussed in Section 4.

If there are $m$ links need to be improved or built and each of the links has $\bar{r}$ candidate projects in average, there may be $2^{m\bar{r}}$ potential combinations of investment strategies. Thus the computational complexity of the upper-level programming model is $O(2^{m\bar{r}})$. This means that the quantity of potential solutions increases exponentially as the number of candidate links in the network grows.

Each of the upper-level solutions represents a scheme of railway network planning. For each scheme, freight flow should be assigned to the corresponding network to estimate the integrated benefit. Therefore, when the upper-level generates a solution, it will call the lower-level model and assign the freight flow on the network. Then the result of lower-level will be substituted into the upper-level to calculate the objective function. Since there are $2^{m\bar{r}}$ potential solutions of upper-level programming, the lower-level programming which focuses on RFFA problem may be executed $2^{m\bar{r}}$ times. The RFFA problem is usually formulated as linear programming model or non-linear programming model. If the lower-level programming is a linear one, its computational complexity should be polynomial. For example, the complexity of a node-arc based

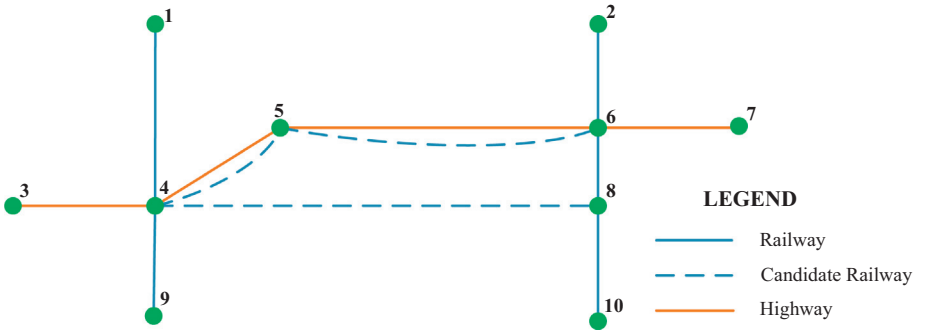



**Fig. 2.** Illustration of the trade-off between investment and carbon emissions.

model with continuous decision variables is $O(n^k)$ (Here $n$ represents the number of the nodes on the network and $k$ is a const which depends on the form of the decision variables). Thus the complexity of the whole model is $O(2^{m'}n^k) \approx O(2^{m'})$. If the lower-level model is a non-linear one, the computational complexity will be much higher and the model will be more difficult to solve.

## 4. Lower-Level programming of RNDP

In order to solve L0 in detail, three typical flow assignment approaches are analyzed in this section, including RFFA based on tree-shaped path, RFFA based on arc-path, and RFFA based on node-arc.

### *4.1. Flow assignment based on tree-shaped path*

In railway freight operation management, when two or more freight flows to the same destination merge at a railway yard, although they may have different origins, they will be considered as one flow and run through the same path during the remainder of the trip. This consideration forms a unique pattern of the flow paths, which resembles a tree at the macro level. This pattern is defined as the tree-shaped path (Lin, 1997).

The decision variable of the flow assignment based on tree-shaped path is denoted as $x_{ij}^k$, which means the volume of freight flow that from $i$ to $j$, and chooses the adjacent station $k$ as the first front station.

The principle of the tree-shaped path is illustrated in Fig. 3 (Lin, 1997).

In Fig. 3, the three purple dotted lines represent the flows $f_{17}$, $f_{27}$ and $f_{37}$. The actual flow from ③ to ⑦ (denoted as $x_{37}$) can be calculated as follows:

$$x_{37} = x_{37}^4 + x_{37}^6 = f_{17} + f_{27} + f_{37} = f_{37} + x_{17}^3 + x_{27}^3 \tag{20}$$

In the railway operation, the flow $x_{37}$ is treated as a single flow, which can choose either path ③→④→⑤→⑦ or ③→⑥→⑦. In other words, flow $x_{37}$ can choose either ④ or ⑥ as its first front station. This situation can be formulated as follows:

$$x_{37}^4 x_{37}^6 = 0 \tag{21}$$

Thus, the flow assignment based on the tree-shaped path is formulated as follows:

$$(\text{L1}) \max Z_L^1 = \sum_{i \in V} \sum_{j \in V} \beta_{ij} f_{ij} - \sum_{m \in E} \sum_{q \in Q_m} c_m^q l_m \omega_m y_m^q \tag{22}$$

$$\text{s.t.} \quad f_{ij} + u_{ij} = F_{ij} \quad \forall i \in V, j \in V, i \neq j \tag{23}$$

$$\sum_{k \in V(i)} x_{ij}^k - \sum_{h \in V(i)} x_{hj}^i = f_{ij} \quad \forall i \in V, j \in V, i \neq j \tag{24}$$

$$\sum_{k \in V(i)} \sum_{h \in V(i), h \neq k} x_{ij}^k x_{ij}^h = 0 \quad \forall i \in V, j \in V, i \neq j \tag{25}$$

$$\omega_{m'} = \omega_{m''} \leqslant \frac{1}{2} \sum_{q \in Q_m} A_m^q y_m^q \quad \forall m \in E \tag{26}$$

$$x_{ij}^k \geqslant 0 \quad \forall i \in V, j \in V, i \neq j, k \in V(i) \tag{27}$$

where $\omega_{m'}$ and $\omega_{m''}$ are the volume of freight flow on arc $m' = (u, v)$, and $m'' = (v, u)$, respectively. They are calculated as follows:

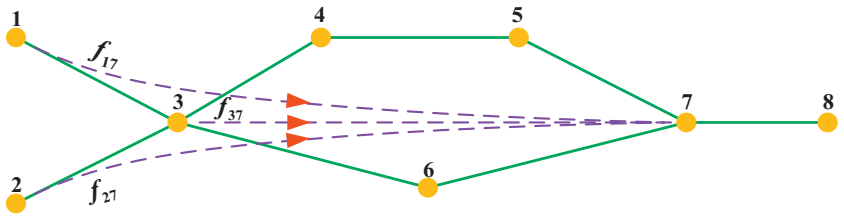



**Fig. 3.** Schematic diagram of merger of freight flow.

$$\omega_{m'} = \sum_{t \in V} x_{ut}^{v} \tag{28}$$

$$\omega_{m''} = \sum_{t \in V} x_{vt}^{u} \tag{29}$$

In L1, the objective function contains two items: the first one is the total income of the railway company and the second one is the total cost. Thus, the objective function aims to maximize the total profit of transporting freight. L1 contains five constraints: (1) the value of an O-D pair equals the sum of feasible flow and infeasible flow; (2) flow conservation; (3) each flow can only choose one station as its first front station; (4) the capacity constraint of each link; and (5) the decision variable should be non-negative. According to L1, we can obtain the flow assignment that conforms to the feature of the tree-shaped path.

### 4.2. Flow assignment based on arc-path

The flow assignment based on tree-shaped path contains many quadratic terms which will make the computational complexity increase significantly. In fact, in the network design stage, an approximate flow assignment from the macro level is enough. Thus, the constraint of tree-shaped path can be omitted. Kuby et al. (2001) adopted arc-path based model which is a linearized one in the railway network design problem. The core idea of arc-path based model is selecting appropriate paths from the set of candidate paths while meeting the objective of the model. Here we improved the model proposed by Kuby and formulate L2 as follows.

$$(\mathrm{L2})\ \max Z_L^2 = \sum_{i \in V} \sum_{j \in V} \beta_{ij} f_{ij} - \sum_{m \in E} \sum_{q \in Q_m} c_m^q l_m \omega_m y_m^q \tag{30}$$

$$\text{s.t.} \quad f_{ij} + u_{ij} = F_{ij} \quad \forall i \in V,\ j \in V,\ i \neq j \tag{31}$$

$$\sum_{p \in P_{ij}} x_{ij}^{p} = f_{ij} \quad \forall i \in V,\ j \in V,\ i \neq j \tag{32}$$

$$\omega_{m'} = \omega_{m''} \leqslant \frac{1}{2} \sum_{q \in Q_m} A_m^q y_m^q \quad \forall m \in E \tag{33}$$

$$0 \leqslant x_{ij}^{p} \leqslant f_{ij} \quad \forall i \in V,\ j \in V,\ i \neq j,\ p \in P_{ij} \tag{34}$$

where $\omega_{m'}$ and $\omega_{m''}$ are calculated as follows:

$$\omega_{m'} = \sum_{i \in V} \sum_{j \in V,\ j \neq i} \sum_{p \in P_{ij}} \delta_{ij}^{pm'} x_{ij}^{p} \tag{35}$$

$$\omega_{m''} = \sum_{i \in V} \sum_{j \in V,\ j \neq i} \sum_{p \in P_{ij}} \delta_{ij}^{pm''} x_{ij}^{p} \tag{36}$$

In L2, $P_{ij}$ represents the set of candidate paths between $i$ and $j$; $\delta_{ij}^{pn}$ is 0-1 parameter which is defined as follows:

$$\delta_{ij}^{pn} = \begin{cases} 1 & \text{if arc } n \text{ is on the path } p \text{ from } i \text{ to } j \\ 0 & \text{otherwise} \end{cases}$$

$x_{ij}^{p}$ is the decision variable, which means the volume of freight flow from $i$ to $j$, and goes through path $p$.

The objective function of L2 also aims to maximize the total profit of transporting freight. It contains four constraints: (1) the sum of feasible flow and infeasible flow is constant; (2) the volume of feasible flow equals the sum of flows on all the paths; (3) capacity constraint; and (4) the value range of the decision variable. The advantage of L2 is that it can take

specified paths[5] into consideration. If some shipments in the railway network must go along specified paths, L2 is suitable for assigning the flow.

### 4.3. Flow assignment based on node-arc

Regarding network design problems, using the flow assignment based on arc-path also leads to difficulties. As the L2 describes, each shipment has a set of candidate paths. The candidate paths are influenced by the network's structure. When the decision variables of the upper-level change, the network structure will change. Thus, the paths in the sets should be set again. This will increase the preprocessing workload significantly. If none or few of the shipments need to go along specified paths, the following flow assignment based on node-arc is better.

$$(\mathrm{L3})\ \max Z_L^3 = \sum_{i \in V} \sum_{j \in V} \beta_{ij} f_{ij} - \sum_{m \in E} \sum_{q \in Q_m} c_m^q l_m \omega_m y_m^q \tag{37}$$

$$\text{s.t.}\quad f_{ij} + u_{ij} = F_{ij} \quad \forall i \in V,\ j \in V,\ i \neq j \tag{38}$$

$$\sum_{n \in O_v} x_{ij}^n - \sum_{n \in I_v} x_{ij}^n = \begin{cases} f_{ij} & v = i \\ 0 & \text{otherwise} \\ -f_{ij} & v = j,\ j \neq i \end{cases} \quad \forall v \in V \tag{39}$$

$$\omega_{m'} = \omega_{m''} \leqslant \frac{1}{2} \sum_{q \in Q_m} A_m^q y_m^q \quad \forall m \in E \tag{40}$$

$$0 \leqslant x_{ij}^n \leqslant f_{ij} \quad \forall i \in V,\ j \in V,\ i \neq j,\ n \in G \tag{41}$$

where $\omega_{m'}$ and $\omega_{m''}$ are calculated as follows:

$$\omega_{m'} = \sum_{i \in V} \sum_{j \in V,\ j \neq i} x_{ij}^{m'} \tag{42}$$

$$\omega_{m''} = \sum_{i \in V} \sum_{j \in V,\ j \neq i} x_{ij}^{m''} \tag{43}$$

In L3, $I_v$ and $O_v$ represent the sets of arcs entering and emanating from node $v$ respectively. $x_{ij}^n$ is the decision variable, which means the volume of freight flow from $i$ to $j$, and goes through arc $n$ .

The objective function of L3 also aims to maximize the total profit of transporting freight. It contains four constraints: (1) the flow conservation of each O-D pair; (2) the flow conservation of each node; (3) capacity constraint; and (4) the value range of the decision variable. L3 is a linear programming model that requires little preprocessing. Adopting this model can reduce the difficulty in solving the problem significantly.

## 5. Practical considerations

### 5.1. Passenger flow

Not including the high-speed railway, most railways serve both passenger trains and freight trains. The passenger trains also occupy the railway capacity. Thus, the influence of passenger flows should be considered. Since the path of freight flows are planned by the railway company from the perspective of system optimization, the freight flow can be assigned by using the models described in this paper. However, the passenger flow assignment is quite different. Passengers have different preferences, making their judging criteria for choosing paths different. For example, two passengers from the same origin who are traveling to the same destination may choose different paths. This leads to it improper to assign passenger flow with the freight flow assignment model. In China, passenger trains take precedence over freight trains. Therefore, the reasonable way to address the problem is to assign the passenger flows with the Logit model, deduct the capacity that is occupied by the passenger flow first, and then solve the freight flow assignment model with the constraint of the remaining capacity.

### 5.2. Empty car flow

In railway freight transportation, both the loaded and empty car flows will occupy the capacity of railway. In China, the percentage of empty cars' travel distance is 35.27% out of the total travel distance of all cars.[6] Thus, when estimating the traffic demand, both cars should be taken into consideration. Empty cars are usually distributed according to the surplus

[5] In China, for some reason, some shipments must go along specified paths rather than the shortest path.

[6] The data is cited from the Compilation of National Railway Statistics of China (2015).

and shortage of each type of empty cars at each station. If a station will have more loaded cars than unloaded cars, empty cars are demanded. Otherwise, the station can supply empty cars. Obviously, the empty car distribution problem in the NDP framework is very complex. Without loss of generality, we use the following basic model which aims at minimizing the total travel distance of empty cars.

$$\min Z = \sum_{i \in V} \sum_{j \in V} l_{ij} \xi_{ij} \tag{44}$$

$$\text{s.t.} \quad \sum_{j \in V} \xi_{ij} = S_i \quad \forall i \in V \tag{45}$$

$$\sum_{i \in V} \xi_{ij} = D_j \quad \forall j \in V \tag{46}$$

$$\xi_{ij} \geqslant 0 \quad \forall i \in V,\ j \in V,\ i \neq j \tag{47}$$

In the model, $\xi_{ij}$ is the decision variable, which represents the quantity of cars from $i$ to $j$. $S_i$ and $D_j$ mean the supply of empty cars in station $i$ and demand of empty cars in station $j$, respectively. The travel distance of delivering empty cars from $i$ to $j$ is denoted by $l_{ij}$ .

Therefore, the process of estimating the O-D matrix, including empty car flow, can be divided into the following six steps:

| | |
|---|---|
| Step 1: | Obtain the freight O-D of railway and highway, respectively. |
| Step 2: | Based on the volume of shipments entering and emanating from each station, calculate the difference of value between loaded cars and unloaded cars. |
| Step 3: | Transform the difference of value into supply or demand of empty cars according to average load. |
| Step 4: | Solve the above model to obtain the empty car flows. |
| Step 5: | Transform the empty car flows into equivalent virtual freight flows according to the average load. |
| Step 6: | Add the railway, highway and equivalent virtual freight flows together to obtain the total O-D matrix. |

### 5.3. Relationship among links in a corridor

Since the construction projects usually focus on an entire railway corridor instead of a single segment, a dependency may exist among the projects. For example, in Fig. 4, the railway company is planning to improve the capacity of the railway from A to E. Generally speaking, if a certain project (e.g. double-track electrified railway) is implemented on link 2, then the same project should usually be implemented on links 1, 3 and 4.

From the perspective of mathematical description, this relationship can be formulated as follows:

$$y_1^q = y_2^q = y_3^q = y_4^q \quad \forall q \tag{48}$$

By considering this factor, the number of candidate schemes can be reduced.

## 6. Numerical example

### 6.1. The input data

In this section, a case with 15 nodes is designed to test the model. The network is shown in Fig. 5.

The lengths and capacity of the links are shown in Table 1.

In the network in Fig. 5, links 1–7, 1–2 and 8–15 are single track railways without electrification, and link 6–7 is a candidate railway. Others are double-track electrified railways. The information of the candidate projects are shown in Table 2. We assume that the budget is 40 billion CNY.

In the network in Fig. 5, links 1–2 and 1–7 are in the same corridor. Thus, they require the same project. This can be formulated as follows:

$$y_{(1,2)}^q = y_{(1,7)}^q \quad \forall q \tag{49}$$

Since there is no candidate projects for other links, their corresponding decision variables are all constant. Besides, doing nothing to the link is also deemed to be a project. Therefore, the number of candidate schemes can be calculated as follows:

$$(|Q_{(1,2)}| + 1) \times (|Q_{(6,7)}| + 1) \times (|Q_{(8,15)}| + 1) = (1 + 1) \times (2 + 1) \times (1 + 1) = 12 \tag{50}$$

The OD matrix is shown in Table 3.

According to the method described in Section 5, we calculated the virtual flows, which are shown in Table 4.

Thus, the initial O-D matrix can be modified into the matrix as shown in Table 5.

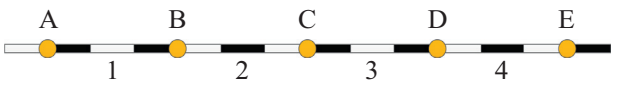



**Fig. 4.** Schematic diagram of dependency among projects.

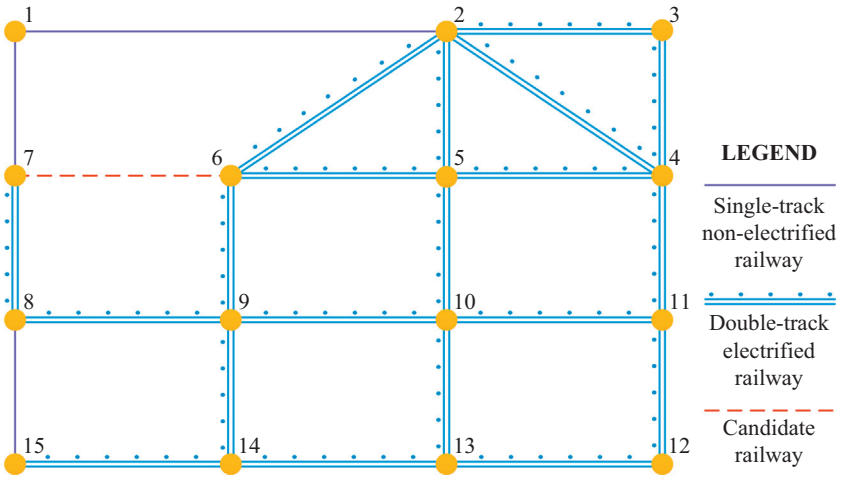



**Fig. 5.** Network for testing the model.

**Table 1**
Length ($l_m$) and initial capacity ($A_m^0$) of the links (kilometer, million ton per year).

| Node 1 | Node 2 | Length | Capacity | Node 1 | Node 2 | Length | Capacity |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 444 | 36 | 6 | 9 | 159 | 160 |
| 1 | 7 | 36 | 36 | 7 | 8 | 88 | 100 |
| 2 | 3 | 299 | 160 | 8 | 9 | 225 | 200 |
| 2 | 4 | 137 | 160 | 8 | 15 | 265 | 36 |
| 2 | 5 | 92 | 160 | 9 | 10 | 132 | 160 |
| 2 | 6 | 122 | 160 | 9 | 14 | 161 | 160 |
| 3 | 4 | 285 | 160 | 10 | 11 | 62 | 160 |
| 4 | 5 | 87 | 100 | 10 | 13 | 152 | 160 |
| 4 | 11 | 239 | 160 | 11 | 12 | 118 | 160 |
| 5 | 6 | 73 | 160 | 12 | 13 | 124 | 100 |
| 5 | 10 | 182 | 160 | 13 | 14 | 159 | 100 |
| 6 | 7 | 280 | 0 | 14 | 15 | 220 | 100 |

*Note:* In the table, Node 1 and Node 2 represent the endpoints of each link.

We assume that the freight which cannot be serviced by train will all be delivered by truck. The distances of highway between the O-D pairs are shown in Table 6.

Unit charges for transporting freight between the nodes are shown in Table 7.

The values of other related parameters are set as follows: $\rho$ = 0.07CNY/ton-km, $c_m^q$ = 0.14CNY/ton-km, $\theta^{\text{Road}}$ = 49.40 g/ton-km, $\theta^{\text{Rail}}$ = 10.53 g/ton-km, $T_m^q$ = 30 years.

### 6.2. Results and discussion

In this case study, we assume that no shipment needs to go along specified path. Thus L3 is chosen as the lower-level model. We use Lingo to solve L3 for the 12 candidate schemes. Based on the freight flow assignment, we calculate the total transportation completed by railway (TTR, billion ton-km), volume of total infeasible flow (TIF, million ton), total transportation completed by highway (TTH, billion ton-km), total carbon emissions (TCE, million ton), average load of each link (AL, million ton), average detour ratio (ADR), the percentage of the total travel distance of empty cars out of the total travel distance of all cars (EDP), and total investment (TI, billion CNY). All of them are shown in Table 8. Because three schemes are outside of the budget, they are not shown in the table.

**Table 2**
Information of candidate projects.

| Links | Projects | | | |
|---|---|---|---|---|
| | Single-track and electrified railway | | Double-track and electrified railway | |
| | $I_m^q$ (billion CNY) | $A_m^q$ (million ton) | $I_m^q$ (billion CNY) | $A_m^q$ (million ton) |
| 1–2 | – | – | 22.2 | 96 |
| 1–7 | – | – | 1.8 | 96 |
| 6–7 | 14.4 | 60 | 24.6 | 180 |
| 8–15 | – | – | 13.2 | 96 |

**Table 3**
The O-D matrix (million ton per year).

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | – | 0.83 | 2.38 | 2.73 | 0.64 | 2.17 | 1.80 | 0.97 | 2.97 | 2.97 | 1.16 | 2.53 | 2.42 | 2.40 | 1.23 |
| 2 | 2.26 | – | 1.84 | 0.78 | 1.58 | 0.55 | 2.60 | 2.96 | 0.73 | 0.88 | 1.03 | 1.25 | 1.44 | 2.97 | 2.32 |
| 3 | 2.32 | 0.91 | – | 1.82 | 0.98 | 0.50 | 0.76 | 0.63 | 0.99 | 2.04 | 0.60 | 2.46 | 2.52 | 2.73 | 1.06 |
| 4 | 2.54 | 2.76 | 2.84 | – | 1.62 | 0.58 | 1.42 | 1.85 | 2.96 | 1.94 | 1.66 | 0.66 | 2.82 | 2.89 | 1.96 |
| 5 | 1.94 | 1.28 | 0.87 | 0.74 | – | 0.73 | 1.18 | 0.98 | 0.81 | 2.33 | 2.70 | 2.11 | 1.31 | 2.69 | 1.99 |
| 6 | 1.98 | 1.64 | 2.35 | 2.68 | 2.84 | – | 1.96 | 2.16 | 2.64 | 1.04 | 1.87 | 1.35 | 0.67 | 2.33 | 0.72 |
| 7 | 2.14 | 2.34 | 36.70 | 2.16 | 1.26 | 0.52 | – | 1.64 | 2.86 | 1.78 | 1.42 | 8.30 | 2.18 | 1.97 | 2.82 |
| 8 | 1.20 | 2.46 | 30.70 | 2.11 | 0.52 | 2.90 | 2.24 | – | 1.29 | 0.78 | 2.40 | 59.00 | 0.92 | 1.78 | 2.52 |
| 9 | 1.75 | 1.76 | 2.20 | 2.35 | 1.54 | 0.63 | 1.39 | 2.16 | – | 1.67 | 2.82 | 0.65 | 2.67 | 1.98 | 0.52 |
| 10 | 1.95 | 2.09 | 2.37 | 1.18 | 2.59 | 2.57 | 2.55 | 2.42 | 1.41 | – | 1.35 | 2.67 | 2.34 | 2.33 | 1.30 |
| 11 | 0.82 | 1.56 | 0.59 | 1.56 | 1.53 | 0.78 | 2.09 | 1.94 | 0.76 | 2.75 | – | 1.40 | 2.35 | 0.52 | 2.71 |
| 12 | 2.27 | 0.91 | 0.63 | 1.88 | 1.66 | 2.30 | 1.46 | 2.82 | 2.18 | 1.87 | 0.76 | – | 2.44 | 0.61 | 2.11 |
| 13 | 2.23 | 0.77 | 1.96 | 1.26 | 1.64 | 2.52 | 1.20 | 1.54 | 1.19 | 2.83 | 1.60 | 1.03 | – | 0.97 | 1.97 |
| 14 | 1.87 | 0.56 | 1.74 | 1.03 | 2.04 | 1.97 | 0.82 | 2.53 | 1.24 | 1.78 | 2.70 | 2.03 | 2.34 | – | 1.60 |
| 15 | 2.03 | 2.78 | 0.64 | 1.20 | 1.57 | 2.17 | 1.64 | 1.84 | 1.86 | 1.65 | 1.88 | 1.89 | 0.80 | 1.16 | – |

**Table 4**
Virtual flows (million ton per year).

| Origin | Destination | Volume | Origin | Destination | Volume |
|---|---|---|---|---|---|
| 1 | 8 | 0.10 | 11 | 8 | 2.59 |
| 3 | 2 | 0.54 | 12 | 8 | 63.43 |
| 3 | 4 | 5.02 | 13 | 8 | 1.70 |
| 3 | 6 | 4.99 | 13 | 10 | 2.81 |
| 3 | 7 | 44.98 | 14 | 8 | 2.88 |
| 3 | 8 | 11.96 | 14 | 9 | 0.20 |
| 5 | 6 | 0.35 | 15 | 8 | 1.72 |

**Table 5**
Modified O-D matrix (million ton per year).

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | – | 0.83 | 2.38 | 2.73 | 0.64 | 2.17 | 1.80 | 1.07 | 2.97 | 2.97 | 1.16 | 2.53 | 2.42 | 2.40 | 1.23 |
| 2 | 2.26 | – | 1.84 | 0.78 | 1.58 | 0.55 | 2.60 | 2.96 | 0.73 | 0.88 | 1.03 | 1.25 | 1.44 | 2.97 | 2.32 |
| 3 | 2.32 | 1.45 | – | 6.84 | 0.98 | 5.49 | 45.74 | 12.59 | 0.99 | 2.04 | 0.60 | 2.46 | 2.52 | 2.73 | 1.06 |
| 4 | 2.54 | 2.76 | 2.84 | – | 1.62 | 0.58 | 1.42 | 1.85 | 2.96 | 1.94 | 1.66 | 0.66 | 2.82 | 2.89 | 1.96 |
| 5 | 1.94 | 1.28 | 0.87 | 0.74 | – | 1.08 | 1.18 | 0.98 | 0.81 | 2.33 | 2.70 | 2.11 | 1.31 | 2.69 | 1.99 |
| 6 | 1.98 | 1.64 | 2.35 | 2.68 | 2.84 | – | 1.96 | 2.16 | 2.64 | 1.04 | 1.87 | 1.35 | 0.67 | 2.33 | 0.72 |
| 7 | 2.14 | 2.34 | 36.70 | 2.16 | 1.26 | 0.52 | – | 1.64 | 2.86 | 1.78 | 1.42 | 8.30 | 2.18 | 1.97 | 2.82 |
| 8 | 1.20 | 2.46 | 30.70 | 2.11 | 0.52 | 2.90 | 2.24 | – | 1.29 | 0.78 | 2.40 | 59.00 | 0.92 | 1.78 | 2.52 |
| 9 | 1.75 | 1.76 | 2.20 | 2.35 | 1.54 | 0.63 | 1.39 | 2.16 | – | 1.67 | 2.82 | 0.65 | 2.67 | 1.98 | 0.52 |
| 10 | 1.95 | 2.09 | 2.37 | 1.18 | 2.59 | 2.57 | 2.55 | 2.42 | 1.41 | – | 1.35 | 2.67 | 2.34 | 2.33 | 1.30 |
| 11 | 0.82 | 1.56 | 0.59 | 1.56 | 1.53 | 0.78 | 2.09 | 4.53 | 0.76 | 2.75 | – | 1.40 | 2.35 | 0.52 | 2.71 |
| 12 | 2.27 | 0.91 | 0.63 | 1.88 | 1.66 | 2.30 | 1.46 | 66.25 | 2.18 | 1.87 | 0.76 | – | 2.44 | 0.61 | 2.11 |
| 13 | 2.23 | 0.77 | 1.96 | 1.26 | 1.64 | 2.52 | 1.20 | 3.24 | 1.19 | 5.64 | 1.60 | 1.03 | – | 0.97 | 1.97 |
| 14 | 1.87 | 0.56 | 1.74 | 1.03 | 2.04 | 1.97 | 0.82 | 5.41 | 1.44 | 1.78 | 2.70 | 2.03 | 2.34 | – | 1.60 |
| 15 | 2.03 | 2.78 | 0.64 | 1.20 | 1.57 | 2.17 | 1.64 | 3.56 | 1.86 | 1.65 | 1.88 | 1.89 | 0.80 | 1.16 | – |

**Table 6**
Distances of highway ($l_{ij}^{Road}$) between the O-D pairs (kilometer).

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | – | 438 | 737 | 476 | 389 | 316 | 36 | 124 | 349 | 481 | 543 | 661 | 633 | 510 | 389 |
| 2 | 438 | – | 299 | 137 | 92 | 122 | 402 | 490 | 281 | 274 | 336 | 454 | 426 | 442 | 662 |
| 3 | 737 | 299 | – | 285 | 372 | 421 | 701 | 789 | 580 | 554 | 524 | 642 | 706 | 741 | 961 |
| 4 | 476 | 137 | 285 | – | 87 | 160 | 440 | 528 | 319 | 269 | 239 | 357 | 421 | 480 | 700 |
| 5 | 389 | 92 | 372 | 87 | – | 73 | 353 | 441 | 232 | 182 | 244 | 362 | 334 | 393 | 613 |
| 6 | 316 | 122 | 421 | 160 | 73 | – | 280 | 368 | 159 | 255 | 317 | 435 | 407 | 320 | 540 |
| 7 | 36 | 402 | 701 | 440 | 353 | 280 | – | 88 | 313 | 445 | 507 | 625 | 597 | 474 | 353 |
| 8 | 124 | 490 | 789 | 528 | 441 | 368 | 88 | – | 225 | 357 | 419 | 537 | 509 | 386 | 265 |
| 9 | 349 | 281 | 580 | 319 | 232 | 159 | 313 | 225 | – | 132 | 194 | 312 | 284 | 161 | 381 |
| 10 | 481 | 274 | 554 | 269 | 182 | 255 | 445 | 357 | 132 | – | 62 | 180 | 152 | 293 | 513 |
| 11 | 543 | 336 | 524 | 239 | 244 | 317 | 507 | 419 | 194 | 62 | – | 118 | 214 | 355 | 575 |
| 12 | 661 | 454 | 642 | 357 | 362 | 435 | 625 | 537 | 312 | 180 | 118 | – | 124 | 283 | 503 |
| 13 | 633 | 426 | 706 | 421 | 334 | 407 | 597 | 509 | 284 | 152 | 214 | 124 | – | 159 | 379 |
| 14 | 510 | 442 | 741 | 480 | 393 | 320 | 474 | 386 | 161 | 293 | 355 | 283 | 159 | – | 220 |
| 15 | 389 | 662 | 961 | 700 | 613 | 540 | 353 | 265 | 381 | 513 | 575 | 503 | 379 | 220 | – |

**Table 7**
Unit charges ($\beta_{ij}$) for transporting freight between the O-D pairs (CNY).

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | – | 92 | 155 | 100 | 82 | 66 | 8 | 26 | 73 | 101 | 114 | 139 | 133 | 107 | 82 |
| 2 | 92 | – | 63 | 29 | 19 | 26 | 84 | 103 | 59 | 58 | 71 | 95 | 89 | 93 | 139 |
| 3 | 155 | 63 | – | 60 | 78 | 88 | 147 | 166 | 122 | 116 | 110 | 135 | 148 | 156 | 202 |
| 4 | 100 | 29 | 60 | – | 18 | 34 | 92 | 111 | 67 | 56 | 50 | 75 | 88 | 101 | 147 |
| 5 | 82 | 19 | 78 | 18 | – | 15 | 74 | 93 | 49 | 38 | 51 | 76 | 70 | 83 | 129 |
| 6 | 66 | 26 | 88 | 34 | 15 | – | 59 | 77 | 33 | 54 | 67 | 91 | 85 | 67 | 113 |
| 7 | 8 | 84 | 147 | 92 | 74 | 59 | – | 18 | 66 | 93 | 106 | 131 | 125 | 100 | 74 |
| 8 | 26 | 103 | 166 | 111 | 93 | 77 | 18 | – | 47 | 75 | 88 | 113 | 107 | 81 | 56 |
| 9 | 73 | 59 | 122 | 67 | 49 | 33 | 66 | 47 | – | 28 | 41 | 66 | 60 | 34 | 80 |
| 10 | 101 | 58 | 116 | 56 | 38 | 54 | 93 | 75 | 28 | – | 13 | 38 | 32 | 62 | 108 |
| 11 | 114 | 71 | 110 | 50 | 51 | 67 | 106 | 88 | 41 | 13 | – | 25 | 45 | 75 | 121 |
| 12 | 139 | 95 | 135 | 75 | 76 | 91 | 131 | 113 | 66 | 38 | 25 | – | 26 | 59 | 106 |
| 13 | 133 | 89 | 148 | 88 | 70 | 85 | 125 | 107 | 60 | 32 | 45 | 26 | – | 33 | 80 |
| 14 | 107 | 93 | 156 | 101 | 83 | 67 | 100 | 81 | 34 | 62 | 75 | 59 | 33 | – | 46 |
| 15 | 82 | 139 | 202 | 147 | 129 | 113 | 74 | 56 | 80 | 108 | 121 | 106 | 80 | 46 | – |

In the upper-level model, $\lambda$ represents the cost coefficient of carbon emissions, reflecting the extent of attention that the related administration pays to air pollution. For example, if $\lambda = 1$, it means that the administration is willing to pay 1 thousand CNY to reduce 1 ton of carbon emissions. The larger $\lambda$ is, the more the administration is willing to pay to reduce air pollution. We substitute the data in Table 8 into the upper-level model and calculate the values of the objective function with different $\lambda$. The result is shown in Fig. 6.

In Fig. 6, each broken-line represents the values of the nine schemes for a given $\lambda$. As $\lambda$ increases, the shape of the broken-line changes significantly. This indicates that the rank of the schemes changes as $\lambda$ varies. When $\lambda \leqslant 1.5$, scheme 1 (All the links keep their situations unchanged) is best. Therefore, if little or no attention is paid to environmental costs, no candidate project is implemented. However, as $\lambda$ increases, the advantage of scheme 1 gradually weakens, while the rank of other schemes also changes. When $\lambda \geqslant 2.0$, scheme 5 replaces scheme 1 as the best one. In this situation, double-track electrified railway should be built between node 6 and 7. When $\lambda \geqslant 6.0$, scheme 1 becomes the worst one. The result indicates that the consideration of carbon emissions will influence the decision. Sometimes the decision may be changed thoroughly. Therefore, based on the current circumstances, the model is of great value to improve the rationality of the railway network design.

We apply this method in the project "Study on Construction of the Rail Network and Freight & Passenger Flow Assignment for the Thirteenth Five-Year Plan of China", and suggest building a railway between Xinzhou and Baoding. In the Medium and long-term railway network planning issued by National Development and Reform Commission on July 13th 2016, this railway had been added to the official planned railway network. The railway between Xinzhou and Baoding will significantly reduce the heavy pressure of transporting coal from the mid-west of China to the east of China. Meanwhile, part of the freight flow will be diverted from the highway to railway, and the carbon emissions from freight transportation will significantly reduce.

**Table 8**
Results of computational experiments.

| No. | Description | TTR | TIF | TTH | TCE | AL | ADR | EDP | TI |
|---|---|---|---|---|---|---|---|---|---|
| 1 | $y^0_{1,2}=y^0_{1,7}=1$<br>$y^0_{6,7}=1$<br>$y^0_{8,15}=1$ | 246.54 | 131.19 | 65.66 | 4.97 | 63.86 | 1.82% | 33.56% | 0.00 |
| 2 | $y^0_{1,2}=y^0_{1,7}=1$<br>$y^0_{6,7}=1$<br>$y^1_{8,15}=1$ | 255.53 | 117.53 | 59.39 | 4.75 | 66.19 | 2.76% | 32.38% | 13.20 |
| 3 | $y^0_{1,2}=y^0_{1,7}=1$<br>$y^1_{6,7}=1$<br>$y^0_{8,15}=1$ | 285.88 | 69.88 | 25.96 | 3.42 | 69.04 | 5.32% | 28.94% | 14.40 |
| 4 | $y^0_{1,2}=y^0_{1,7}=1$<br>$y^1_{6,7}=1$<br>$y^1_{8,15}=1$ | 297.83 | 55.07 | 17.64 | 3.14 | 71.92 | 6.94% | 27.78% | 27.60 |
| 5 | $y^0_{1,2}=y^0_{1,7}=1$<br>$y^2_{6,7}=1$<br>$y^0_{8,15}=1$ | 306.99 | 17.77 | 2.18 | 2.47 | 74.12 | 3.60% | 26.95% | 24.60 |
| 6 | $y^0_{1,2}=y^0_{1,7}=1$<br>$y^2_{6,7}=1$<br>$y^1_{8,15}=1$ | 306.99 | 17.77 | 2.18 | 2.47 | 74.12 | 3.60% | 26.95% | 37.80 |
| 7 | $y^1_{1,2}=y^1_{1,7}=1$<br>$y^0_{6,7}=1$<br>$y^0_{8,15}=1$ | 290.99 | 71.43 | 25.71 | 3.46 | 75.36 | 7.50% | 28.43% | 24.00 |
| 8 | $y^1_{1,2}=y^1_{1,7}=1$<br>$y^0_{6,7}=1$<br>$y^1_{8,15}=1$ | 302.51 | 55.07 | 17.66 | 3.19 | 78.34 | 8.62% | 27.35% | 37.20 |
| 9 | $y^1_{1,2}=y^1_{1,7}=1$<br>$y^1_{6,7}=1$<br>$y^0_{8,15}=1$ | 308.29 | 30.33 | 7.26 | 2.73 | 74.44 | 6.19% | 26.84% | 38.40 |

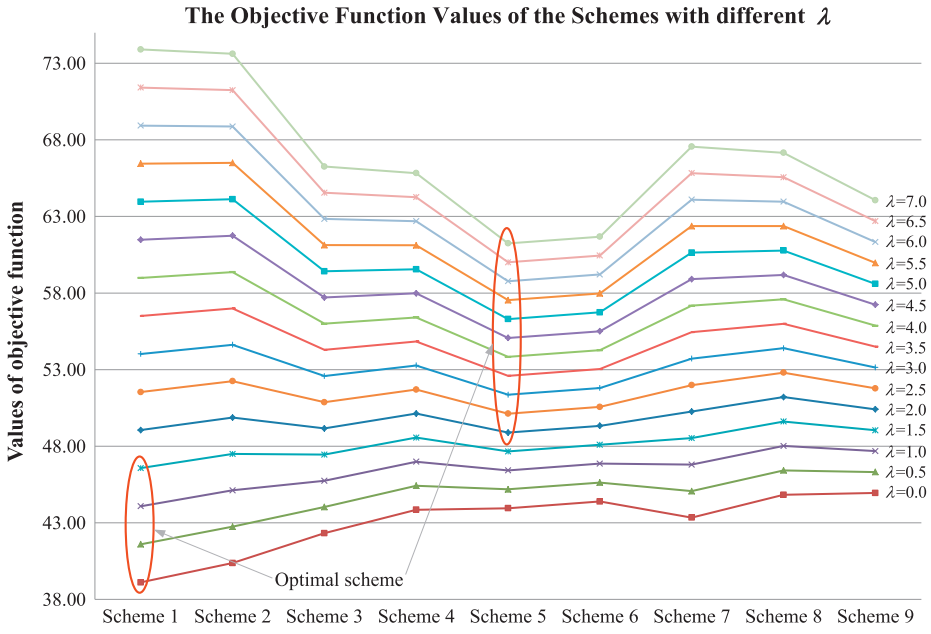



**Fig. 6.** Objective function values of the schemes with different $\lambda$.

## 7. Conclusions

In this paper, we studied the railway network design problem under the background of reducing carbon emissions. First, we analyzed the carbon emissions of highway and railway transportation. Based on the data, we proposed that building a

railway can also reduce carbon emissions of the transportation system. Thus, when designing the railway network, we suggested balancing the environmental costs and the investment. Therefore, we developed a novel 0-1 bi-level programming model. The upper-level is for selecting projects and minimizing the sum of carbon emissions costs, opportunity costs, transportation costs and annual investment. The lower-level is for assigning the freight flow on the network generated by the upper-level and provide the upper-level with data to calculate the objective function. In order to comply with the practice of railway construction and operation, we introduced the methods to deal with passenger flow, empty car flow and dependency among projects. Finally, we conducted a numerical example to test the model and demonstrate the decision-making process when choosing projects. The result indicates that the final decision may be heavily influenced by the consideration of carbon emissions.

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